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Roll No.....

CE-8045 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Design of Prestressed Concrete Structures
(Elective - II)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume missing data if suitably.

1. Classify various method of post tensioning system based on wedge action. Describe any one method of post tensioning in details with sketches.
2. Briefly outline the magnet's method of computing the horizontal and transverse stress in end blocks subjected to concentrated force from anchorage.
3. Define the following:
 - i) Tendon
 - ii) Pretensioning
 - iii) Post tensioning
 - iv) Load balancing
4. a) What are the advantages and disadvantages of prestressed concrete member over RCC member?
b) Write a short note on need of high strength concrete and steel.
5. Explain the analysis of anchorage zone stresses in post tensioned members. How is the bursting tensile force calculated?
6. A post tensioned prestressed beam of rectangular section 250mm wide is to be designed for an imposed load of 12 kN/m, over 12m span. The stress in the concrete must not exceed 17 N/mm^2 in compression and 1.4 N/mm^2 in tension at any time and the loss of prestress may be assumed to be 15%. Calculate the minimum possible depth of beam. Also calculate minimum prestressing force and eccentricity for provided depth.
7. Sketch the layout of tendons of a PSC continuous beam:
 - i) Straight
 - ii) Curved in elevation

OR

How will you improve shear resistance of a P.S.C. beam?

8. Write short notes on any two of the following:
 - a) Principles of prestressing
 - b) End zone reinforcement
 - c) Design of continuous beams
 - d) Design of purlin
 - e) Design of railway sleepers
